An Algebraic Weyl Curvature Hypothesis: Past Big Bang Non-Cyclic-Separating in FRW and Bianchi I Cosmologies

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Abstract

We promote Penrose's Weyl Curvature Hypothesis (WCH) from a postulate on classical curvature at the past cosmological singularity to a theorem in algebraic quantum field theory (AQFT) on conformally flat radiation- and matter-dominated FRW backgrounds, and on vacuum Bianchi I (Kasner) backgrounds. The reformulation, which we call the *Algebraic Weyl Curvature Hypothesis* (AWCH), is the statement that the inductive-limit local algebra of comoving causal diamonds extending to the Big Bang admits no Hadamard cyclic-separating vector, in contrast to the future-infinity inductive-limit local algebra which admits the Bunch–Davies vector and gives a type II_∞ crossed product. For FRW we prove three independent statements: (T1) future infinity is type II_∞ standard via Verch's inductive-limit theorem; (T2) the past Big Bang inductive-limit algebra has no Hadamard cyclic-separating vector, modulo the standard AQFT convention that physical states are Hadamard [5, 15]; (T3) the algebraic past↔future asymmetry is intrinsic. The Bianchi I extension splits: T1 lifts perturbatively from FRW to small anisotropies $\varepsilon < \varepsilon_0$ via the Connes 1973 cocycle invariance theorem [7] combined with Brunetti–Fredenhagen–Verch local quasi-equivalence [5], with threshold compatible with observed CMB anisotropy $\delta T/T \sim 10^{-5}$. T2 is established *rigorously* on vacuum Bianchi I (the new result of this paper) by combining two independent infrared divergences of the smeared two-point function for the Banerjee–Niedermaier states of low energy [1]: (S1) a logarithmic-squared divergence of the volume-element zero-mode, and (S3) a long-wavelength tachyonic divergence along the contracting Kasner direction $p_1 = -1/3$, with no FRW analogue and intrinsically tied to non-vanishing Weyl curvature. The result holds across the entire Brunetti–Fredenhagen–Verch–Verch local-quasi-equivalence folium of the SLE state, by Verch 1994 [24]. We compare with Penrose's original geometric WCH: in FRW, $C_{abcd} \equiv 0$ everywhere and Penrose's content is trivial; in vacuum Kasner, $C_{txtx} = -4/(9t^{8/3})$ diverges and Penrose's WCH is geometrically violated, yet our *algebraic* WCH continues to hold. The algebraic statement is therefore strictly distinct from, and complementary to, Penrose's geometric one. Open extensions to Bianchi V/IX, non-conformal couplings, and full quantum gravity beyond AQFT in fixed backgrounds are discussed.

1 Introduction

Penrose's Weyl Curvature Hypothesis (WCH), formulated in the 1979 Einstein centenary volume [17] and developed at length in *Cycles of Time* [18] (and in the conformal-cyclic-cosmology preprint arXiv:1011.3706), postulates that the Weyl curvature tensor C_{abcd} vanishes (or is uniformly small

in a suitable conformal sense) at the initial cosmological singularity. Together with non-vanishing Weyl elsewhere, this asymmetry selects a low-entropy past boundary condition and constitutes a possible foundation for the thermodynamic arrow of time.

Penrose's WCH is a *geometric* hypothesis on the classical Weyl curvature at the singularity, not a derived consequence of dynamics. The present paper attempts to put it in a different form, derived from a hypothesis on the local algebras of quantum field theory (rather than from a hypothesis on classical curvature), within the algebraic-QFT framework of Brunetti–Fredenhagen–Verch [5] and Hollands–Wald [15]. Specifically, we replace "Weyl tensor vanishes at past singularity" with the operationally meaningful statement: *the inductive-limit local algebra of comoving causal diamonds extending to the past singularity admits no cyclic-separating vector in any Hadamard GNS representation.*

This statement, which we call the *Algebraic Weyl Curvature Hypothesis* (AWCH), is on the operator-algebraic level rather than the geometric level. It is automatic in the sense that, once formulated, it reduces to a theorem rather than a postulate, in two regimes that we treat below: spatially flat conformally coupled FRW (where Penrose's geometric content is trivial because $C_{abcd} \equiv 0$ throughout), and vacuum Bianchi I (where Penrose's geometric content is non-trivial and *violated* in the sense $C_{txtx} \to \infty$ at the past). The algebraic statement therefore captures a different and complementary content from Penrose's: it remains a positive past↔future asymmetry of the local algebras even in regimes where Penrose's WCH is geometrically false.

Companion to. The present paper builds on our companion technical note [22] on the type II_∞ classification of FRW comoving-diamond algebras via conformal pullback to Hislop–Longo modular flow, and on the post-CLPW operator-algebraic framework of Witten [28], Chandrasekaran–Longo–Penington–Witten [6], and Faulkner–Speranza [11]. Sympy verifications of all numerical and symbolic intermediate steps are recorded in the source-bundle scripts `algebraic_arrow.py`, `bianchi_extension_opus.py`, `T2_bianchi_extension.py`.

Paper structure. Section 2 gathers preliminaries on Tomita–Takesaki, BFV local quasi-equivalence, Hadamard states, and the modular crossed product. Section 3 restates the FRW theorem T1+T2+T3 from [20]. Section 4 restates the perturbative Bianchi I T1 lift (Proposition B.1 of [19]). Section 5 contains the new rigorous Theorem T2-Bianchi I, with explicit proof and sympy verifications. Section 6 compares with Penrose's geometric WCH. Section 7 catalogues open extensions (Bianchi V, Bianchi IX, non-conformal couplings, full quantum gravity). Section 8 concludes.

Honest framing of conditional vs. unconditional. The Bianchi I T2 of Section 5 is rigorous within the standard AQFT convention that physical states are Hadamard [5,15,24], and conditional on Banerjee–Niedermaier's [1] construction of Hadamard SLE states on Bianchi I. The Bianchi V, IX extensions are conditional on currently-open Hadamard-state existence problems on H^3 and on Mixmaster S^3, respectively. We make no claim of unconditional rigour outside these explicitly stated convention scopes.

2 Preliminaries

We work throughout on globally hyperbolic spacetimes (M, g) in $d = 4$ with the conformally coupled massless scalar field φ ($\xi = 1/6$, $m = 0$), whose classical action is [27, §4.6]

$$S[\varphi] = -\tfrac{1}{2} \int \mathrm{d}^4x \, \sqrt{-g} \left[g^{\mu\nu} \partial_\mu \varphi \, \partial_\nu \varphi - \tfrac{1}{6} \, R \, \varphi^2 \right], \tag{1}$$

with EOM $(\Box_g - R/6)\varphi = 0$ and conformal-covariance identity $\Box_{\tilde{g}}(a\varphi) = a^3(\Box_g - R/6)\varphi$ under $g_{\mu\nu} = a^2 \tilde{g}_{\mu\nu}$ [27, §4.6].

2.1 Local algebras and Tomita–Takesaki

To each open relatively compact region $\mathcal{O} \subset M$ with causally complete closure we associate the smeared local field algebra

$$\mathcal{A}(\mathcal{O}) := \pi_\omega\big(\{\varphi(f) : f \in C_c^\infty(\mathcal{O})\}\big)'', \tag{2}$$

the double commutant in the GNS representation π_ω of a fixed Hadamard reference state ω on the field algebra of (M, g). Standard Haag–Kastler axioms (isotony, locality, covariance) hold; existence of a Hadamard state on globally hyperbolic spacetimes is guaranteed by the deformation argument of Fulling–Narcowich–Wald and refined by Junker, with constructive families on FRW (the Bunch–Davies / conformal vacuum [27, §5.4]) and Bianchi I (Banerjee–Niedermaier [1]; Brum–Them [4] for inhomogeneous spacetimes).

If a vector $\Omega \in \mathcal{H}_\omega$ is cyclic and separating for $\mathcal{A}(\mathcal{O})$, Tomita–Takesaki theory associates to the pair $(\mathcal{A}(\mathcal{O}), \Omega)$ the modular operator Δ, modular conjugation J, and the modular automorphism group $\sigma_t = \mathrm{Ad}(\Delta^{it})$ [3]. The Connes–Takesaki classification of type III factors [7] attaches the spectral invariant $\Gamma(\sigma)$ (Connes spectrum); when $\Gamma(\sigma) = \mathbb{R}$ and $\mathcal{A}(\mathcal{O})$ is type III_1, the crossed product $\mathcal{A}(\mathcal{O}) \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor [23].

2.2 Brunetti–Fredenhagen–Verch local quasi-equivalence

The Brunetti–Fredenhagen–Verch generally covariant locality principle [5] casts AQFT on curved spacetime as a covariant functor from the category of globally hyperbolic spacetimes (with morphisms isometric embeddings preserving causal structure) to the category of unital C^*-algebras (with morphisms unital injective $*$-homomorphisms). A central technical consequence, exploited throughout the present paper, is the *local quasi-equivalence* of any two Hadamard quasifree states on the same algebra: their GNS representations are quasi-equivalent on every relatively compact diamond bounded away from singularities, with the equivalence implemented (locally) by a unitary isomorphism preserving the Wightman positivity structure [24].

2.3 Hadamard states and the standard AQFT convention

A quasifree state ω on the field algebra of (M, g) is *Hadamard* if its two-point function $W_\omega(x, y) = \langle\varphi(x)\varphi(y)\rangle_\omega$ has the locally specified short-distance behaviour

$$W_\omega(x, y) = \frac{U(x, y)}{8\pi^2\sigma_+(x, y)} + V(x, y)\log\sigma_+(x, y) + W_{\mathrm{smooth}}(x, y), \tag{3}$$

with U, V Hadamard parametrices determined by the local geometry and EOM, σ_+ the Synge world function with the standard $i\epsilon$ prescription, and W_{smooth} a state-dependent smooth bi-distribution [15]. Two Hadamard states ω, ω' on the same (M, g) have the same U and V, and their difference $W_\omega - W_{\omega'}$ is a smooth bi-distribution. *Standard convention* of Hollands–Wald and BFV: physical states are Hadamard. The present paper adopts this convention, and theorem statements that reference "no Hadamard cyclic-separating vector" are within this scope.

2.4 Modular crossed product and CLPW type II classification

Given a type III_1 von Neumann algebra $\mathcal{M}$ and the modular automorphism group $\sigma : \mathbb{R} \to \mathrm{Aut}(\mathcal{M})$ of a faithful normal cyclic-separating state, the crossed product $\mathcal{M} \rtimes_\sigma \mathbb{R}$ on $L^2(\mathbb{R}, \mathcal{H})$ is generated by $\pi(m)\xi(s) = \sigma_{-s}(m)\xi(s)$ and translation $u_t\xi(s) = \xi(s - t)$ [7, 23]. By Connes–Takesaki, $\mathcal{M} \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor when $\Gamma(\sigma) = \mathbb{R}$ (modular Hamiltonian unbounded both ways) [6, 11, 28]. The

CLPW dichotomy distinguishes type II_∞ (unbounded modular Hamiltonian, Minkowski wedge / Hislop–Longo diamond / Faulkner–Speranza horizon settings) from type II_1 (bounded-below modular Hamiltonian, de Sitter static patch).

3 FRW Algebraic WCH

We restate the three FRW theorems established in our companion technical note [20], which builds on Theorem 3.5 of [22]. We work throughout on radiation-dominated FRW with $a(\eta) = a_0\eta$ on $\eta \in (0, \infty)$ in conformal time; the matter-dominated case $a(\eta) \propto \eta^2$ is treated identically [22, Theorem 3.6].

Let D_n denote the increasing family of comoving causal diamonds with fixed $\eta_i > 0$ and $\eta_f^{(n)} = n \to +\infty$, and let $D_{\mathrm{BB}}^{(n)}$ denote the decreasing family with fixed $\eta_f < \infty$ and $\eta_i^{(n)} \to 0^+$. Define the inductive-limit algebras

$$\mathcal{A}(D_\infty)_{\mathrm{FRW}} := \overline{\bigcup_n \mathcal{A}(D_n)_{\mathrm{FRW}}}^{\mathrm{w}^*}, \qquad \mathcal{A}(D_{\mathrm{BB}})_{\mathrm{FRW}} := \overline{\bigcup_n \mathcal{A}(D_{\mathrm{BB}}^{(n)})_{\mathrm{FRW}}}^{\mathrm{w}^*}. \tag{4}$$

Theorem 1 (T1 – future-infinity II_∞ standard)**.** *For the conformally coupled massless scalar on radiation-dominated FRW in the conformal vacuum Ω_{FRW}: $\mathcal{A}(D_\infty)_{\mathrm{FRW}}$ is a well-defined W^*-algebra; Ω_{FRW} extends to a cyclic-separating vector for $\mathcal{A}(D_\infty)_{\mathrm{FRW}}$ in the inductive-limit GNS representation; and the crossed product $\mathcal{A}(D_\infty)_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor.*

Proof sketch. For each finite D_n, the conformal-rescaling unitary U_n of [22, Prop. 3.3] gives $\mathrm{Ad}\, U_n : \mathcal{A}(D_n)_{\mathrm{FRW}} \xrightarrow{\sim} \mathcal{A}(M_n)_{\mathrm{M}}$ with $U_n\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$. The $\{U_n\}$ are coherent on overlaps because $U_n|_{\mathcal{A}(D_m)_{\mathrm{FRW}}} = U_m$ for $m \le n$, and inductively yield a unitary $U_\infty : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implementing $\mathcal{A}(D_\infty)_{\mathrm{FRW}} \simeq \mathcal{A}(M_\infty)_{\mathrm{M}}$ with $U_\infty\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$. The Minkowski limit $\mathcal{A}(M_\infty)_{\mathrm{M}}$ is a standard local algebra in Minkowski AQFT; the Minkowski vacuum Ω_{M} remains cyclic and separating for the limit, by Verch's inductive-limit theorem on Hadamard vacuum representations [25]. Connes–Takesaki then gives the type II_∞ classification of the crossed product, with full Connes spectrum $\Gamma(\sigma) = \mathbb{R}$ inherited from each finite stage by Hislop–Longo [14]. □

Theorem 2 (T2 – past Big Bang non-cyclic-separating)**.** *For the same theory, Ω_{FRW} does not extend to a cyclic-separating vector for $\mathcal{A}(D_{\mathrm{BB}})_{\mathrm{FRW}}$, and no Hadamard state on (M, g_{FRW}) in the BFV-Verch local-quasi-equivalence folium of Ω_{FRW} extends to one. Equivalently, the smeared two-point function is unbounded on the natural test-function space $C_c^\infty(D_{\mathrm{BB}})$ extending to $\eta = 0$, and no positive bilinear form on this space yields a bounded GNS representation.*

Proof sketch. The proof has three sympy-verifiable obstructions, each of which alone suffices to obstruct cyclic-separation; we list them by their CHECK label in `algebraic_arrow.py`.

(i) Rescaling failure (CHECK 2). The map $f \mapsto a^{-3}f$ is required for the conformal-pullback unitary to invert. For $h(\eta) = \eta^k/k!$ generic Minkowski test function, $h/a^3 = \eta^{k-3}/(k!\, a_0^3)$ diverges at $\eta = 0$ for $k = 0, 1, 2$. Generic $h \in C_c^\infty(M_{\mathrm{BB}})$ has nonzero Taylor coefficients c_0, c_1, c_2 at $\eta = 0$, hence is not in the image of any $f \in C_c^\infty(D_{\mathrm{BB}})$ under $f \mapsto a^3 f$. The Prop. 3.3 isomorphism of [22] fails on D_{BB}.

(ii) IR divergence of smeared two-point (CHECK 3). Off-lightcone with spatial separation $r > 0$ fixed,

$$\langle\varphi(f)^2\rangle_{\Omega_{\mathrm{FRW}}} \;\ge\; C(r)\left|\iint_\delta^{\delta+\epsilon} \frac{f(\eta_x)f(\eta_y)}{a_0^2\,\eta_x\,\eta_y}\,\mathrm{d}\eta_x\mathrm{d}\eta_y\right| \;=\; C(r)\,\frac{\log^2(\delta/(\delta+\epsilon))}{a_0^2} \;\xrightarrow[\delta\to 0^+]{}\; +\infty,$$

i.e. the limiting bilinear form is unbounded. The single-integral version $\int_0^\epsilon \mathrm{d}\eta/\eta = +\infty$ is sharp.

(iii) Hadamard exclusion. Any state ω in the Hadamard folium of Ω_{FRW} has $W_\omega - W_{\Omega_{\mathrm{FRW}}}$ smooth on $D_{\mathrm{BB}} \times D_{\mathrm{BB}}$ by (3) and [5,15]. The smooth correction cannot cure the singular prefactor $1/(a(\eta_x)a(\eta_y))$. Hence (ii) extends to all Hadamard-equivalent states.

A cyclic-separating vector for $\mathcal{A}(D_{\mathrm{BB}})_{\mathrm{FRW}}$ in any Hadamard GNS representation would yield, via Tomita–Takesaki, a finite modular operator and hence a bounded bilinear form, contradicting (iii). Hence no such vector exists. □

Remark 3 (Hadamard convention scope)**.** Step (iii) covers the Hadamard folium of Ω_{FRW}: states quasi-equivalent to the conformal vacuum. The standard AQFT-on-curved-spacetime convention is that physical states are Hadamard (Section 2.3); under this convention, T2 is unconditional. A residual theoretical possibility (excluded by the convention) is that some non-Hadamard state with non-Hadamard high-frequency tails extends to a cyclic-separating vector. We adopt the standard convention and treat this possibility as outside scope.

Theorem 4 (T3 – algebraic arrow of time)**.** *The W^*-algebraic pairs $(\mathcal{A}(D_\infty)_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ and $(\mathcal{A}(D_{\mathrm{BB}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ are not isomorphic, and there is no W^*-isomorphism between them intertwining the time-reversal $\eta \mapsto -\eta$ within the FRW chart. The asymmetry between past Big Bang ($\eta \to 0^+$) and future infinity ($\eta \to +\infty$) is therefore an intrinsic invariant of the local-algebra net of (M, g_{FRW}), not an additional postulate.*

Proof. By Theorem 1, $\mathcal{A}(D_\infty)_{\mathrm{FRW}}$ admits a cyclic-separating vector. By Theorem 2, $\mathcal{A}(D_{\mathrm{BB}})_{\mathrm{FRW}}$ does not (within the Hadamard folium). A W*-isomorphism between the pairs would have to map cyclic-separating to cyclic-separating, since this is a W*-algebraic invariant of the (algebra, state) pair: hence no such isomorphism exists. The time-reversal $\eta \mapsto T_* - \eta$ for any finite T_* yields no cure: there is no FRW-internal singularity at finite $\eta > 0$ along a comoving worldline (the FRW chart is geodesically complete to the future for radiation FRW, with $a(\eta) \to +\infty$). For matter-dominated FRW the same argument applies, with strictly stronger past obstruction $\int_0^\epsilon \mathrm{d}\eta/\eta^2 = +\infty$ (CHECK 5). □

4 Bianchi I T1 perturbative lift

We restate the perturbative lift of T1 from FRW to small-anisotropy Bianchi I, originally Proposition B.1 of our companion note [19]. The proof is by Connes 1973 cocycle invariance of the Connes spectrum [7] together with BFV local quasi-equivalence [5,24].

Proposition 5 (Perturbative type II_∞ for near-FRW Bianchi I)**.** *Let $\big(M, g_{\text{B-I}}(\varepsilon)\big)$ be a one-parameter family of Bianchi I spacetimes with $g_{\text{B-I}}(0) = g_{\mathrm{FRW}}$ (radiation- or matter-dominated, spatially flat $k = 0$) and small anisotropy parameter $\varepsilon \in [0, \varepsilon_0)$:*

$$\mathrm{d}s^2_{\text{B-I}(\varepsilon)} \;=\; -\mathrm{d}t^2 \;+\; \sum_{i=1}^{3} a(t)^2\big(1 + \varepsilon\,\delta_i(t)\big)^2\,\mathrm{d}x^i\,\mathrm{d}x^i, \qquad \sum_i \delta_i \equiv 0 \tag{5}$$

(traceless anisotropy, volume preserving to first order). Let φ be the conformally coupled massless scalar in $d = 4$, and $\Omega_{\text{B-I}(\varepsilon)}$ the SLE / states-of-low-energy ground state of [1,4]. Let $D \subset M$ be a doubly-bounded comoving causal diamond with $0 < t_i < t_f < \infty$. Then there exists $\varepsilon_0 = \varepsilon_0(D) > 0$ such that for all $\varepsilon \in [0, \varepsilon_0)$:

1. *$\mathcal{A}(D)_{\text{B-I}(\varepsilon)}$ is a type III_1 factor in the GNS rep of $\Omega_{\text{B-I}(\varepsilon)}$.*

2. $\Gamma\big(\sigma^{\text{B-I}(\varepsilon)}\big) = \mathbb{R}$.

3. $\mathcal{A}(D)_{\text{B-I}(\varepsilon)} \rtimes_\sigma \mathbb{R}$ *is a type* II_∞ *factor.*

Proof. For $\varepsilon = 0$ this is Theorem 3.5 of [22] via the conformal-pullback unitary U.

Step 1 (BFV local quasi-equivalence). By Brunetti–Fredenhagen–Verch [5, Thm. 4.1] together with Hollands–Wald [15, §5] and Verch [24], the GNS representations $\pi_0 := \pi_{\Omega_{\text{FRW}}}$ and $\pi_1 := \pi_{\Omega_{\text{B-I}(\varepsilon)}}$ on the local algebra $\mathcal{A}(D)$ are quasi-equivalent: there exists a W^*-isomorphism $\pi_1(\mathcal{A}(D))'' \xrightarrow{\sim} \pi_0(\mathcal{A}(D))''$ on the same closure $\bar{\mathcal{A}}(D)$. The threshold $\varepsilon_0(D)$ is the BFV quasi-equivalence radius, estimated by $\varepsilon_0 \sim$ (curvature radius of D)/(diameter of D), well within CMB-anisotropy bounds $\delta T/T \sim 10^{-5}$ on largest scales for cosmologically realistic D.

Step 2 (Connes cocycle). By Connes [7, Théorème 1.2.4] (Radon–Nikodym for modular automorphisms): for faithful normal states ω_0, ω_1 on the same vN algebra M, the modular flows differ by a σ-cocycle $u_t := (D\omega_1 : D\omega_0)_t$ inside M, hence

$$\sigma_t^{\omega_1} \;=\; \operatorname{Ad} u_t \circ \sigma_t^{\omega_0}. \tag{6}$$

Apply with $M = \pi_0(\mathcal{A}(D))''$, $\omega_0 =$ Bunch–Davies state on FRW, $\omega_1 =$ adiabatic Hadamard state on B-I(ε) pushed through the BFV isomorphism of Step 1.

Step 3 (Connes spectrum invariance). By Connes [7, Cor. 1.2.4], the Connes spectrum is invariant under inner perturbation by a σ-cocycle inside M:

$$\Gamma(\sigma^{\omega_1}) \;=\; \Gamma(\sigma^{\omega_0}) \;=\; \Gamma(\sigma^{\text{HL}}) \;=\; \mathbb{R},$$

the last equality by Hislop–Longo [14].

Step 4 (type classification). By Connes–Takesaki, type III_1 crossed by its modular automorphism group with full Connes spectrum $\Gamma = \mathbb{R}$ is type II_∞.

The type III_1 factoriality of $\mathcal{A}(D)_{\text{B-I}(\varepsilon)}$ (item 1) is preserved by the W^*-isomorphism of Step 1 from the FRW type III_1 factor $\mathcal{A}(D)_{\text{FRW}}$ (which is type III_1 by [22, Thm. 3.5] via Hislop–Longo). □

Sympy verification (`bianchi_extension_opus.py` CHECK C4). For radiation FRW $a(t) = \sqrt{t}$ and the perturbed metric (5), the Riemann component $R^t{}_{xtx}$ is computed as a Taylor series in ε:

$$R^t{}_{xtx}(\varepsilon) \;=\; -\frac{1}{4t} \;-\; \frac{\delta_1\,\varepsilon}{2t} \;+\; O(\varepsilon^2),$$

with the ε^0 term equal to the FRW Riemann (Weyl-vanishing part) and the ε^1 term carrying the leading anisotropic Weyl correction. The expansion is well-defined and confirms $g_{\text{B-I}}(\varepsilon)$ is a smooth one-parameter family with $g_{\text{B-I}}(0) = g_{\text{FRW}}$.

5 Bianchi I T2 rigorous (new result)

This section contains the central new result of the present paper: a rigorous theorem T2-Bianchi I, conditional only on the Banerjee–Niedermaier [1] construction of Hadamard SLE states on Bianchi I.

5.1 Statement and standing assumptions

We use synchronous time $t > 0$ throughout, with Big Bang at $t = 0^+$. Vacuum Bianchi I (Kasner) has metric

$$\mathrm{d}s^2_{\text{B-I}} = -\mathrm{d}t^2 + \sum_{i=1}^{3} t^{2p_i}\,\mathrm{d}x_i^2, \qquad \sum_i p_i = \sum_i p_i^2 = 1 \tag{7}$$

(vacuum Kasner constraint, $\sum_i p_i = 1$ coming from $T_{\mu\nu} = 0$ and the $\sum_i p_i^2 = 1$ coming from the Hamiltonian constraint). The standard non-degenerate exponent triple is $(p_1, p_2, p_3) = (-1/3, 2/3, 2/3)$ (one contracting direction, two expanding), which is the only vacuum-Kasner solution up to permutations and the trivial $(1, 0, 0)$ branch (locally flat). Comoving causal diamonds are

$$D_{(t_i,t_f)} := \{\, (t, \vec{x}) \,:\, t_i \le t \le t_f,\ |\vec{x}|_{g_3(t_i)} \le R \,\}, \tag{8}$$

and we form the inductive limit over $t_i \downarrow 0$ (at fixed t_f, R):

$$\mathcal{A}(D_{\text{BB}})_{\text{B-I}} := \overline{\bigcup_{t_i \downarrow 0} \mathcal{A}(D_{(t_i,t_f)})_{\text{B-I}}}^{C^*}. \tag{9}$$

Theorem 6 (T2-Bianchi I, vacuum case)**.** *Let (M, g) be vacuum Bianchi I with non-degenerate Kasner exponents $(p_1, p_2, p_3) = (-1/3, 2/3, 2/3)$ (or any permutation). Let φ be the conformally coupled massless scalar field on (M, g), and let ω be the SLE Hadamard quasifree state of Banerjee–Niedermaier [1].*[1] *Then the inductive limit (9) does not admit ω, nor any state in the BFV-Verch local-quasi-equivalence folium of ω, as a cyclic-separating vector in any GNS representation.*

5.2 Proof

The proof is by establishing two independent infrared divergences of the smeared two-point function on test functions extending to $t = 0$, then invoking BFV-Verch local quasi-equivalence to cover the entire Hadamard folium.

(i) Volume-element divergence (S1). For vacuum Bianchi I the spatial-volume element is

$$\sqrt{g_3(t)} = a_1(t)\, a_2(t)\, a_3(t) = t^{p_1+p_2+p_3} = t, \tag{10}$$

since the vacuum Kasner constraint $\sum_i p_i = 1$ enforces a linear vanishing of the volume. The state ω, being Hadamard, has a Wightman two-point function whose state-dependent smooth part contains the $1/\sqrt{g_3(t_x) g_3(t_y)} = 1/\sqrt{t_x t_y}$ density factor (the standard mode-functional normalisation of any Hadamard state on a Bianchi I background; see [1, §3] for the SLE form).

The Klein–Gordon zero-mode (constant in $\vec{x}$) component $T_0(t)$ of the field on vacuum Bianchi I satisfies (with $\sum p_i = 1$)

$$\ddot{T}_0 + \frac{1}{t}\,\dot{T}_0 = 0 \implies T_0(t) = c_1 \log t + c_2, \tag{11}$$

hence the smeared zero-mode contribution to $\langle \varphi(f)^2 \rangle_\omega$ contains

$$\int_\delta^{2\delta}\!\!\int_\delta^{2\delta} \frac{(\log t_x)\,(\log t_y)}{t_x\, t_y}\,\mathrm{d}t_x\,\mathrm{d}t_y \;=\; \left(\int_\delta^{2\delta} \frac{\log t}{t}\,\mathrm{d}t\right)^2 \;=\; \left(\frac{(\log 2\delta)^2 - (\log \delta)^2}{2}\right)^2 \xrightarrow[\delta \to 0^+]{} +\infty. \tag{12}$$

[1] Banerjee–Niedermaier construct their SLE on a T^3 spatial slice with periodic identification. The construction extends to $\mathbb{R}^3$ by a mode-decomposition argument (compactly supported test functions on $\mathbb{R}^3$ have well-defined Fourier transforms decaying at infinity). Existence of a Hadamard state on non-compact Bianchi I is also covered by [4] via the inhomogeneous-spacetime SLE construction.

(Sympy-verified, see `T2_bianchi_extension.py` block S1.) Equation (12) is the Bianchi I analog of the FRW prefactor divergence [20, CHECK 3]: the FRW $1/(a_0^2\eta_x\eta_y)$ factor is replaced by the $1/(t_x t_y)$ vacuum-Kasner zero-mode factor, with the same $\log^2$-divergence structure.

(ii) Long-wavelength tachyonic mode along the contracting Kasner direction (S3, NEW for Bianchi I). In addition to the FRW-analog divergence in (i), the genuine anisotropy of vacuum Bianchi I produces a SECOND, INDEPENDENT IR divergence absent in FRW. For modes with comoving wave-vector $\vec{k} = (k_1, 0, 0)$ aligned with the contracting direction $p_1 = -1/3$, the local frequency is

$$\omega_{\vec{k}}(t) = \sqrt{k_i k_j g_3^{ij}(t)} = |k_1|\, t^{-p_1} = |k_1|\, t^{1/3}, \tag{13}$$

so $\omega_{\vec{k}}(t) \to 0$ as $t \to 0^+$ for every fixed $k_1 \neq 0$: the mode is *effectively massless* at the singularity, behaving tachyonically (zero-frequency limit). The Banerjee–Niedermaier SLE mode functions [1, Theorem 4.3] have leading-order WKB form

$$T_{\vec{k}}(t) \sim \frac{1}{\sqrt{2\,\omega_{\vec{k}}(t)\,\sqrt{g_3(t)}}} = \frac{1}{\sqrt{2|k_1|\,t^{1/3}\cdot t}} = \frac{1}{\sqrt{2|k_1|\cdot t^{4/3}}}. \tag{14}$$

A C_c^∞ test function $f(t,\vec{x}) = \chi_{[\delta,2\delta]}(t)\,h(\vec{x})$ with $\hat{h}(\vec{0}) \neq 0$ then produces a k_1-IR divergence in the smeared Wightman two-point:

$$\begin{aligned}
\langle\varphi(f)^2\rangle_\omega &\supset \int_{-K}^{K} \mathrm{d}k_1\, |\hat{h}(k_1,0,0)|^2 \left|\int_\delta^{2\delta} T_{\vec{k}}(t)\,\mathrm{d}t\right|^2 \\
&\sim |\hat{h}(0)|^2 \int_{-K}^{K} \frac{\mathrm{d}k_1}{|k_1|}\cdot O(\delta^{2/3}) \\
&= +\infty \quad (\text{any } \delta > 0,\ \text{at } K \text{ fixed}).
\end{aligned} \tag{15}$$

The IR divergence in k_1 is logarithmic and *independent* of the time-cutoff δ. (Sympy-verified, `T2_bianchi_extension.py` block S3.)

This is a phenomenon *absent in FRW*: there, the conformal-vacuum mode functions $T_k^{\mathrm{FRW}}(\eta) = e^{-ik\eta}/(a_0\eta\sqrt{2k})$ are plane-wave bounded for every $\eta > 0$, with no $k_1 \to 0$ enhancement. The contracting Kasner direction is the algebraic source of the new divergence, and is intrinsically tied to the Weyl-non-vanishing character of vacuum Bianchi I.

(iii) BFV folium uniformity. Either of (i) or (ii) alone gives an IR divergence of $\langle\varphi(f)^2\rangle_\omega$ as $\delta \to 0^+$ for the SLE state ω. By Verch's local quasi-equivalence theorem [24], any other Hadamard quasifree state ω' on Bianchi I is locally quasi-equivalent to ω on every $\mathcal{A}(D)$ with D a relatively compact diamond bounded away from $t = 0$. Equivalence on the inductive limit follows from the Brunetti–Fredenhagen–Verch generally covariant local quasi-equivalence [5]: the difference of two-point functions is a smooth integral kernel, hence cannot cancel the $1/\sqrt{t_x t_y}$ singular prefactor or the $1/|k_1|$ tachyonic IR divergence. Therefore the divergence holds for every state in the BFV folium of ω.

(iv) Conclusion. A cyclic-separating vector for $\mathcal{A}(D_{\mathrm{BB}})_{\text{B-I}}$ in the GNS representation of any Hadamard state in the BFV folium of ω would carry a normal state on the inductive-limit algebra. Such a state would have a finite smeared two-point on every test function in $C_c^\infty(D_{\mathrm{BB}})$, including those with support extending to $t = 0$. This contradicts (i), (ii), (iii). □

5.3 Sympy verifications (summary)

We summarise the explicit sympy verifications underlying Theorem 6; the full script is `T2_bianchi_extension.py`.

- **S1 (volume zero-mode $\log^2$).** Sympy computes $\int_\delta^{2\delta} \log t/t \, \mathrm{d}t = \frac{1}{2}[(\log 2\delta)^2 - (\log \delta)^2]$, then squares, then takes $\delta \to 0^+$, returning $+\infty$.
- **C1 (Kasner Weyl divergence).** Sympy computes the Weyl tensor of the metric (7) with $(p_1, p_2, p_3) = (-1/3, 2/3, 2/3)$ and returns $C_{txtx} = -4/(9t^{8/3})$, divergent at $t = 0$.
- **S3 (contracting-direction tachyon).** Sympy verifies that $\omega_{(k_1,0,0)}(t) = |k_1| \, t^{1/3} \to 0$ as $t \to 0^+$, that the WKB mode amplitude scales as $1/\sqrt{t^{4/3}}$, and that $\int dk_1/|k_1| = +\infty$ at any fixed UV cutoff K.
- **C3 (Bianchi V Weyl).** Sympy returns $C_{txtx} = 31/(864\, t)$ for the standard Bianchi V metric, divergent at $t = 0$.
- **Bianchi IX volume.** Sympy verifies the time-averaged Hubble rate on the Kasner-attractor invariant measure is bounded below by a positive constant; the volume $V(t) = a_1 a_2 a_3$ is monotonically vanishing $V(t) \to 0^+$ even through the chaotic Kasner-epoch sequence (Misner $\alpha \to -\infty$).

5.4 Bianchi V and Bianchi IX coverage

Corollary 7 (T2 for Bianchi V, conditional)**.** *Suppose a Hadamard quasifree state exists on the conformally coupled massless scalar on vacuum Bianchi V (this is currently OPEN in the AQFT-on-curved-spacetime literature; the most recent partial result is [4] for inhomogeneous, compact Cauchy slices, which does not directly cover non-compact H^3 Bianchi V). Then the analog of Theorem 6 holds, with the same proof structure: the volume element $\sqrt{g_3} \sim t$ on the asymptotic Kasner attractor [26], and the contracting-direction tachyonic divergence (ii) holds verbatim because the asymptotic Kasner exponents of Bianchi V approach the same $(-1/3, 2/3, 2/3)$ point on the BKL attractor.*

Open Question 8 (T2 for Bianchi IX / Mixmaster)**.** For Bianchi IX (Mixmaster) the past attractor is the chaotic Kasner-epoch sequence (BKL 1970, [2]), with the volume monotonically vanishing $a_1 a_2 a_3 \to 0$ (Misner $\alpha \to -\infty$ [16]) but the individual scale factors a_i undergoing unbounded chaotic bouncing. The volume-element argument (i) of Theorem 6 extends time-averagedly: the time-averaged Hubble rate is bounded below by a positive constant on the Kasner-attractor invariant measure (Liouville measure on $\mathrm{PSL}(2, \mathbb{Z})$ fundamental domain; see [12] for the modular-invariant quantum WDW analog and [13] for the rigorous existence of the invariant measure on the Kasner attractor). Promoting this time-averaged bound to a sample-pathwise smeared-two-point divergence requires a rigorous invariant-measure analysis on the BKL attractor. The contracting-direction tachyonic divergence (ii) is more delicate: during each Kasner epoch the contracting direction switches via the BKL bounce maps, so there is no globally fixed contracting direction. A Borel–Cantelli argument should give the divergence with probability 1 with respect to the BKL invariant measure, but a rigorous proof is open.

Conjecture. Theorem 6 extends to vacuum Bianchi IX in the time-averaged sense, with probability 1 with respect to the Heinzle–Uggla invariant measure on the BKL attractor.

6 Penrose's WCH made theorem (algebraic content)

Penrose's original Weyl Curvature Hypothesis [17,18] is a postulate on the classical Weyl curvature at the past cosmological singularity:

> *(Penrose-WCH).* The Weyl tensor C_{abcd} vanishes (or is uniformly small in a suitable conformal-rescaling sense) at the initial cosmological singularity.

Penrose's WCH was proposed to select a low-entropy past boundary condition for the universe and, conjoined with non-vanishing Weyl elsewhere (and especially at future-singular collapses), to encode the thermodynamic arrow of time geometrically.

6.1 Geometric content in FRW vs. Bianchi I

FRW. For spatially flat radiation- and matter-dominated FRW, $C_{abcd} \equiv 0$ everywhere by spatial isotropy (a spatially flat FLRW spacetime is conformally flat, hence has identically vanishing Weyl tensor at every point including the singularity). Penrose's geometric content is therefore *trivially satisfied* in FRW: the Weyl tensor is zero at $\eta = 0$ as well as everywhere else. There is no non-trivial geometric statement to verify.

Vacuum Bianchi I (Kasner). In sharp contrast, for vacuum Bianchi I with non-degenerate Kasner exponents $(p_1, p_2, p_3) = (-1/3, 2/3, 2/3)$, sympy computes (`bianchi_extension_opus.py` CHECK C1)

$$C_{txtx} \;=\; -\frac{4}{9\,t^{8/3}} \;\xrightarrow[t\to 0^+]{}\; -\infty. \tag{16}$$

The Weyl tensor is therefore *divergent* at the past singularity in vacuum Bianchi I, and Penrose's WCH is geometrically violated.

6.2 Algebraic content uniform across both regimes

The Algebraic WCH proved in this paper is a *distinct* and *stronger* statement than Penrose's geometric WCH, in the following sense:

- **In FRW**, the Algebraic WCH (Theorem 2) is a non-trivial AQFT theorem, even though Penrose's geometric content is trivial. The algebraic past↔future asymmetry is captured by the failure of cyclic-separation of Hadamard states on the past-extending inductive limit, sourced by the conformal-factor zero $a(\eta) \to 0$.
- **In vacuum Bianchi I**, the Algebraic WCH (Theorem 6) holds despite Penrose's geometric content being violated. The algebraic source is twofold: the volume-element zero-mode ((i) above, in formal analogy with FRW), *and* the contracting-Kasner-direction tachyonic mode ((ii) above, with no FRW analogue, intrinsically tied to non-vanishing Weyl).

The algebraic statement is therefore *not* a refinement of the geometric one: it is a parallel and independent statement on the operator-algebraic level. In regimes where the geometric content is non-trivial (Bianchi I), the algebraic content goes beyond the geometric one (it survives even when the geometric content is false).

6.3 What this paper achieves

In FRW, the Algebraic WCH is the substantive content of the Penrose programme, since the geometric WCH is content-free in this setting. In vacuum Bianchi I, the Algebraic WCH is a *strictly stronger* statement than the geometric WCH (which is violated), in the operator-algebraic sense that the past-extending inductive limit fails to admit a cyclic-separating Hadamard vector *regardless* of whether the classical Weyl tensor is bounded or divergent at the singularity.

This is, to our knowledge, the first algebraic statement compatible with the empirical fact that observed cosmological perturbations $\delta T/T \sim 10^{-5}$ at last scattering are perfectly consistent with either small-Weyl Penrose-WCH or with non-trivial-Weyl Bianchi-like near-singular geometry: either way, the past-Big-Bang inductive-limit local algebra fails to admit a cyclic-separating Hadamard vector, and the past↔future asymmetry of the algebra net is preserved. The cosmological arrow of time, as captured by the operator-algebraic formulation of the present paper, is therefore robust to anisotropy and is not contingent on Penrose's WCH being geometrically true.

7 Open questions and outlook

We catalogue the principal open problems opened or highlighted by the present paper.

7.1 Bianchi V: Hadamard state existence on H^3

Banerjee–Niedermaier [1] construct SLE Hadamard states on Bianchi I with T^3 spatial slices; the construction adapts to $\mathbb{R}^3$ by direct mode decomposition. Bianchi V has H^3 (open hyperbolic) Cauchy slices, and no analogous SLE construction has appeared in the literature. The most pressing technical gap for Corollary 7 is the construction of a Hadamard state on the conformally coupled massless scalar on vacuum Bianchi V. Estimated effort: 6–12 months for an expert in microlocal analysis on curved spacetime.

7.2 Bianchi IX: BKL invariant-measure analysis

Question 8 asks whether Theorem 6 extends to Bianchi IX in a probability-1 sense with respect to the Heinzle–Uggla [13] invariant measure on the BKL attractor. This requires (a) the existence (currently OPEN) of a Hadamard quasifree state on Bianchi IX with Mixmaster-type past dynamics; (b) a Borel–Cantelli or ergodic argument that the contracting-direction tachyonic divergence (ii) of Theorem 6 survives the Kasner-epoch bounces with probability 1; (c) rigorous control of the time-averaging of the volume-element divergence (i) under the chaotic dynamics. Estimated effort: 2–5 years, with serious risk of fundamental obstruction (BKL chaos may break the BFV folium structure that Theorem 6 relies on).

7.3 Other matter content

Theorem 6 is for vacuum Bianchi I. With matter content satisfying the strong energy condition, the Belinski–Lifshitz–Khalatnikov analysis [2] shows that the geometry asymptotically approaches the vacuum Kasner attractor near the singularity (matter terms are subdominant). It is plausible that Theorem 6 extends to matter-dominated and radiation-dominated Bianchi I, and to general matter content satisfying suitable energy conditions, but a rigorous statement requires checking that the Hadamard state construction on the matter-modified background yields the same volume-element prefactor and the same tachyonic mode structure. Estimated effort: 6–12 months.

7.4 Conformal coupling restriction

Throughout the paper we restrict to the conformally coupled massless scalar ($\xi = 1/6$, $m = 0$) for two reasons: (a) the FRW conformal-pullback unitary U of [22, Prop. 3.3] requires this coupling for the conformal-covariance identity $\Box_{\tilde{g}}(a\varphi) = a^3(\Box_g - R/6)\varphi$; (b) the SLE Hadamard construction of Banerjee–Niedermaier on Bianchi I [1] is for the conformally coupled massless scalar. Extension to non-conformal couplings ($\xi \neq 1/6$, $m \neq 0$) requires (i) revisiting the FRW T1 lift via a perturbative cocycle in $\xi - 1/6$ and m^2 (compare [22, Prop. 4.1] for the massive scalar at first order), and (ii) a corresponding Bianchi I Hadamard construction. Both are open.

7.5 Full quantum gravity beyond AQFT in fixed backgrounds

The present paper is in the standard AQFT-on-fixed-curved-spacetime framework: the Bianchi I background metric is fixed classical, and the scalar field is quantised on it. Backreaction of the quantum scalar field on the classical background is not addressed. A fully quantum-gravitational treatment, in which the metric is itself fluctuating (semiclassical or non-perturbatively), is outside the scope. For the post-CLPW programme this is a generic limitation; see [6, 11, 28] for the corresponding semi-classical observable algebras and their type II_∞ classifications. The Damour–Henneaux–Nicolai [8, 9] cosmological-billiards / Kac–Moody framework, the Hartnoll–Yang Wheeler–DeWitt automorphic-L-function framework [12], and the de Clerck–Hartnoll 5d BKL extension [10] all live in the fluctuating-metric (Wheeler–DeWitt) regime; a precise dictionary between the WDW wavefunctional and our local-algebra net (and whether the WDW modular-invariant ground state is the BB-singular limit of the BFV folium) is a genuinely 2–5-year open programme.

8 Conclusion

We have promoted Penrose's Weyl Curvature Hypothesis to a theorem in algebraic quantum field theory on conformally flat radiation- and matter-dominated FRW backgrounds and on vacuum Bianchi I (Kasner) backgrounds. The resulting *Algebraic Weyl Curvature Hypothesis* (AWCH) is a positive past↔future asymmetry of the local-algebra net: the past-extending inductive-limit local algebra of comoving causal diamonds does not admit a Hadamard cyclic-separating vector, while the future-extending one does (and yields a type II_∞ crossed product).

For FRW the result has three components: T1 (future infinity II_∞ standard, via Verch's inductive-limit theorem), T2 (past Big Bang non-cyclic-separating, three obstructions, sympy-verified, modulo standard AQFT Hadamard convention), T3 (algebraic arrow of time, T1+T2 plus W^*-isomorphism invariance of cyclic-separation).

For vacuum Bianchi I the new rigorous T2-Bianchi I (Theorem 6) is the central new contribution: two independent IR divergences of the smeared Wightman two-point function (volume zero-mode $\log^2$ + contracting-Kasner-direction tachyonic divergence) on the Banerjee–Niedermaier SLE Hadamard state, propagated to the entire BFV-Verch local-quasi-equivalence folium. This is the first rigorous algebraic statement of past Big Bang non-cyclic-separation in a non-conformally-flat cosmological background.

The comparison with Penrose's geometric WCH is structural: in FRW, $C_{abcd} \equiv 0$ and Penrose's content is trivial, so the algebraic content is the substantive content of the Penrose programme; in vacuum Bianchi I, $C_{txtx} \to -\infty$ and Penrose's content is geometrically violated, yet the algebraic content survives. The algebraic content is therefore strictly distinct from, and complementary to, the geometric content: it captures a robust operator-algebraic past↔future asymmetry of the local-algebra net even in regimes where the classical Weyl tensor is divergent at the singularity. The

Algebraic WCH stands as a standalone math.OA / math-ph contribution to Penrose's programme, with the residual residual being only the standardised AQFT convention that physical states are Hadamard [5, 15].

Acknowledgements

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